UPI and India's Digital Payments Revolution

Introduction to UPI

UPI stands for Unified Payments Interface. It is an instant, real-time payment system developed by the National Payments Corporation of India (NPCI) that allows people to transfer money between bank accounts instantly using a smartphone, 24 hours a day, 7 days a week, including holidays. With UPI, a person can pay a shopkeeper, send money to a friend, pay bills or collect money with just a few taps, a QR scan or a mobile number, without entering bank account numbers or IFSC codes. Multiple bank accounts can be linked to a single UPI app. UPI has transformed how Indians pay, from big shopping malls to tea stalls and roadside vegetable vendors, and has become a model for the world.

About NPCI

The National Payments Corporation of India is an umbrella organisation for retail payments in India, set up in 2008 by the Reserve Bank of India and the Indian Banks' Association under the Payment and Settlement Systems Act, 2007, as a not-for-profit company. Besides UPI, it runs other systems such as IMPS (Immediate Payment Service), RuPay (India's domestic card network, launched in 2012), NACH, FASTag, Bharat BillPay and the National Financial Switch for ATMs.

When Was UPI Launched and by Whom?

UPI was launched on 11 April 2016 in Mumbai by Dr. Raghuram Rajan, who was the Governor of the Reserve Bank of India at that time. It was developed by NPCI, with the technical support of the RBI, as a pilot with 21 banks and then went fully live on 25 August 2016. The government's BHIM (Bharat Interface for Money) app, built on UPI, was launched by Prime Minister Narendra Modi on 30 December 2016, and was named after Dr. B. R. Ambedkar. After the demonetisation of high-value notes in November 2016, interest in digital payments rose sharply. UPI built on the existing IMPS infrastructure and on the "India Stack", the digital public infrastructure that includes Aadhaar identity, e-KYC and mobile phones, and the Jan Dhan bank accounts.

How UPI Works

Each user creates a Virtual Payment Address (VPA), also called a UPI ID, such as name@bankname, which links to a bank account, and sets a secret UPI PIN of four or six digits. To pay, the user enters the receiver's UPI ID, mobile number or scans a QR code, enters the amount and approves the payment with the UPI PIN. The UPI app sends a request to the sender's bank, which debits the account and sends the amount to the receiver's bank through the NPCI system, in a few seconds. The system supports both person-to-person (P2P) transfers and person-to-merchant (P2M) payments, along with "collect requests", auto-pay for subscriptions, IPO applications and mandates. Popular UPI apps include PhonePe, Google Pay, Paytm, BHIM, Amazon Pay, WhatsApp Pay and apps by banks. UPI Lite allows small payments without a PIN, and UPI 123PAY lets feature-phone users pay through voice calls and missed calls. Typical limits are Rs. 1 lakh per transaction for most payments, with higher limits for certain categories such as tax payments and hospitals.

How Is UPI Different From a Debit Card?

A debit card is a physical plastic card linked to a bank account, with a card number, expiry date and CVV, which is used to withdraw cash at ATMs and to pay at shops through a point-of-sale (POS) machine or online, using a PIN or OTP. It works through card networks such as Visa, Mastercard and RuPay, and the merchant needs a card terminal, and the merchant pays a fee, called the merchant discount rate. UPI, on the other hand, is not a card but a mobile-based payment system that sends money directly from one bank account to another, in real time, with no card, no card details to share and no POS machine, only a smartphone and a QR code. Debit cards require carrying the card, whereas UPI needs only the phone. UPI has no merchant discount rate for most payments, since the government decided on zero charges for UPI payments from January 2020, making it cheap for small businesses. UPI can also transfer money to friends, while a debit card cannot directly send money to another person. Both are linked to bank accounts, and both are secured by a PIN.

Growth and Scale

UPI has grown explosively. From about 0.1 billion transactions a year in 2016-17, it crossed 10 billion transactions in a month in October 2023 and 20 billion transactions in a month in August 2025, with a value of about Rs. 25 lakh crore a month. UPI accounts for about 80 percent or more of India's digital payment transactions by volume and nearly half of all real-time digital payments in the world, according to global studies. India overtook other countries in real-time payments.

UPI Around the World

UPI has been linked with other countries' payment systems. Singapore's PayNow was linked in February 2023, and UPI is accepted in the UAE, Sri Lanka, Mauritius, Nepal, Bhutan, France, where tourists can pay at the Eiffel Tower, and several other countries. NPCI International Payments Limited (NIPL) leads these efforts. Tourists from other countries can use UPI in India.

UPI in Tamil Nadu and Daily Life

In Chennai and across Tamil Nadu, small vendors, auto-rickshaw drivers, tea shops, flower sellers and temple counters accept payments by QR code. Chennai Metro and city buses also support UPI-based ticketing. Many of the world's UPI start-ups and fintech companies have set up teams in Chennai, and Zoho and other companies provide payment solutions. UPI has brought millions of small merchants into the formal financial system.

Safety Tips

UPI is secure but users must be careful: never share the UPI PIN or OTP with anyone, remember that the PIN is needed only to send money and not to receive it, so a "collect request" or QR code to "receive" money is a scam, verify the name of the receiver before paying, use apps only from official stores, beware of fake customer care numbers, and report fraud on the helpline 1930 or at cybercrime.gov.in and to the bank.

Challenges and Future

Challenges include fraud, outages, concentration of the market in a few apps, a cap on market share set at 30 percent for each app, and sustainable funding. The future includes credit on UPI, UPI for offline payments, voice payments in Indian languages, digital rupee integration and more global links.

Quick Facts

- UPI stands for Unified Payments Interface, developed by NPCI.
- UPI was launched on 11 April 2016 by RBI Governor Raghuram Rajan.
- UPI is an instant account-to-account transfer system without a card or POS machine; a debit card is a physical card linked to an account.
- UPI crossed 20 billion monthly transactions in August 2025.
- Report UPI fraud on helpline 1930.